Database Project

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1. Project Overview & Motivation

Within the field of IR (International Relations) there is huge hurdle in the logistics of data, the term "Data Silo" problem. As Artificial Intelligence (AI) has augmented its outreach to redefine military capabilities and become an integral part of how states use it beyond just military in the daily lives of people , information on the trade and transfer for these type of technologies and autonomous systems is super fragmented. Firstly, data on AI is so novel that we don't even have many observations about it. At the moment, a researcher like me who is attempting to track the proliferation of AI technology and its impact on human rights must navigate through a very disorganized landscape of unstructured PDFs or datasets from non-governmental organizations, various government registries from their website, scattered news reports or use economic trade data as a proxy to see and find trends and patterns within huge global trade datasets. . This fragmentation and scattering feature of the data that predominantly a huge part of my research interest and dissertation makes longitudinal analysis nearly impossible and creates so many barriers for fledgling researchers such as me to analyze the global impact of these technologies.

With this research gap or rather void of crisis, I develop the AI Tech Trade Database (AITTD) in this project for my database systems class/ information management class. This project is not just about data collection; rather it's like creating a one spot destination for all the possible needs (ideally, still working on this to make it better) by establishing a centralized, relational system that will allow researchers to track the flow of autonomous systems and military AI or any AI related to any correlation with state surveillance with as much precision as I possibly can. The goal is that by integrating these disparate data sources into a single, unified environment, the AITTD aims to ease any frustration scholars in this interest might have. I am trying to fill the hollow of the "lost" information into a searchable and actionable asset. This system will aim to serve as a foundational tool in order to understand how emerging technologies are being distributed globally, what those trends look like and which geopolitical actors or states are driving the current race of surveillance technology. .

The primary objective of this project is to shift the standard of research in this domain from flat spreadsheets to a dynamic, relational system, which was the most important and key takeaway from an information management class that equipped us to learn the whole idea of the the connection between variables, which is the relational nature and how that is used for the creation of a structured database that retrieves information efficiently. Traditional spreadsheets fail to capture the complex, multi-dimensional relationships inherent in the line of research where we want to account for many relevant variables within our studies. For example, a single country may act as a supplier for multiple different AI systems to various recipients across different years. A relational database handles these one-to-many and also accounts for the many-to-many relationships with ease, ensuring that the data remains organized and accessible even as the complexity of the research grows or the size of the database, which is something highly unlikely to achieve in an excel sheet.

2. Theoretical Database Design & Normalization

Initially this database was part of a causal study paper that I wanted to write, however, as the class progressed and the more I learnt about databases I realized that this database can have utility beyond the limitation of just running a causal or correlational study. The design of the AITTD is now more a side project of my PhD journey and also a way in which I deal with the data collection of a concept that is no new in the current times that we don't really have enough time or years of data on, which is AI. I identified four core entity types: Nations, AI Systems, Trade Agreements, and Manufacturers. Nations provide the geopolitical context, including human rights and democracy scores; AI Systems define the technical nature and autonomy level of the hardware; Manufacturers provide insight into the corporate origins of the tech; and the Trade Agreements act as the link that records the actual movement of technology between states.

The most important component of this database design was the transition from First Normal Form (1NF) to Third Normal Form (3NF). This was part of the class lectures of normalization, and after the couple of classes we had on this I realized how applicable the concept was to my design methods actually. Basically in many initial data sets information is stored in a way that creates massive, a huge amount of redundancy for instance, repeating a manufacturer's name and headquarters address every time a sale of an AI tech is recorded is useless in some ways.

The transition from First Normal Form (1NF) to Third Normal Form (3NF) represented a critical evolution in the architectural integrity of this project. While 1NF ensures that the data is atomic and unique (which was part of the most important concept that was covered in multiple classes ACID), this might simply end up being a flat spreadsheet structure where descriptive attributes of the key variables or as they are called entities are repeated across multiple observations and rows (also called as tuples in the context of database systems) such as a country's regime type or a technology's lethality risk will be are repeated across every single transaction, leading to massive data redundancy/repetition and an increased risk of update anomalies. What does that mean? That means that if there is change in one small feature of suppose a country there is a fear that in the case of a spreadsheet it's likely that I might make more errors by making sure that that small information is updated for every observation where the state that witnessed the small change is updated.

By transitioning therefore to 3NF, I decouple these entities into discrete, logical tables, ensuring that every non-key attribute depends strictly on the primary key. This normalization process eliminated transitive dependencies, meaning that geopolitical or technical information is now stored as a single source of truth. Consequently, a systemic update, such as a shift in a nation's GDP or a reclassification of any attributes information, is reflected across all 750 transfer records instantaneously through relational joins, instead of literally requiring thousands of manual, error-prone edits. By moving to 3NF, I ensured that each piece of information is stored in only one place. Another example of this is the "Manufacturer" attribute. If the manufacturer were simply a text column in the Transfers table, it would lead to redundancy and "update anomalies." Therefore this is great because by doing the normalization I am placing manufacturers in their own separate table, we ensure that if a company changes its headquarters, we update it once in the master table rather than in thousands of individual trade records.

The logic of the system is visually shown in the Entity-Relationship (ER) model. This model relies on the strict use of Primary Keys and Foreign Keys to maintain order. Every trade is uniquely identified by a Primary Key, ensuring that no two records are duplicated. Meanwhile, Foreign Keys act as the "connectors" between tables. For example, the "Recipient Country" column in our Transfers table is a Foreign Key that links directly to the master "Nations" table. This structure ensures that the database physically prevents the entry of a trade for a country that hasn't been officially defined in the system, maintaining the integrity of the entire dataset.

So based on the schema above, I created three tables for now. The four in the schema are, transfers, technology, human rights and countries. The structural backbone of this project is a relational database designed to track the global movement of AI surveillance tools. While conceptually the framework identifies four distinct domains Countries, Technologies, Transfers, and Human Rights, when I was actually building the tables in pgadmin in PostgreSQL I utilized only three tables for efficient optimization of data in my database design. This reduction is not a loss of data in my eyes, but I think instead it's a strategic normalization choice.

In the pgAdmin environment, the data is organized into the Countries, Technologies, and Transfers tables. The Countries table serves as the geographical and political anchor, using `country_id` as its primary key to store stable attributes like regime type and GDP per capita. The Technologies table acts as a technical catalog, using `tech_id` to define equipment specifications such as lethality risk and autonomy levels. Finally, the Transfers table serves as the central "fact table," capturing each specific event of a technology moving to a country under the primary key `transfer_id`. On purpose I chose to embed human rights data . This was specifically because physical integrity and censorship indices can directly be put into the Transfers table instead of having a whole other table in my opinion. By doing so I treat human rights metrics as transactional outcomes of a transfer within a specific given year, allowing the database to capture the reality as much as possible at the exact moment a technology was acquired. If human rights were stored in separate, static tables like excel sheets, it would be so difficult to track how those scores fluctuate year-by-year in direct response to new AI acquisitions. Now within the dashboard, all human rights information remains fully accessible because it is retrieved through the Transfers table and this is enabling a seamless analytical link between a country's identity and its subsequent rights performance.

To ensure cohesiveness within the database design system rather than a collection of isolated lists, the design logic relies on a rigorous system of referential integrity. Each table utilizes a unique primary key `country_id`, `tech_id`, and `transfer_id` to ensure that no two records are the same, and maintains data integrity. Now these tables are connected via foreign keys within the Transfers table, which links back to the table of original Countries and Technologies records. I use this relational structure allows the R/Shiny application to use foreign keys to "reconstruct" the full story of a transfer in real-time. By joining these IDs, the system can instantly provide a more nuanced view of the data, informing the user not just that a transfer occurred, but the specific socio-political context, such as the regime type and economic status of the recipient nation. This architecture provides a lean, high-performance environment that scales efficiently as more global data is acquired.

Now for the data, I simulated data for the purposes of this project. Initially, I merged data from different human rights outcomes data sources such as acled, vdem and hrmi for human rights, then AIGS for AI, but then i was having a lot of technical issues with merging the data and thus finally i made the decision to simulate all of the data randomly. I simulated data for 25 countries with 750 transfers. Here, each transfer is the equivalent of a unit of analysis like in political science research.

3. System Architecture & Servers

For the architecture of this project there is a backend and a frontend. The backend is powered by a PostgreSQL Server, which is where the data stays and to visually access this data I used PGAdmin. By housing the data in a dedicated RDBMS, it ensures that the structure of the data remains protected from corruption and that the relationships between tables are strictly enforced by the server's internal logic, instead of relying on the user's spreadsheet habits.

The application layer is the next type of server which is Shiny Server, which acts as the critical bridge between the raw database and the end-user as they host the interactive dashboard part of the project. Shiny utilizes a reactive programming model, meaning the application is constantly "listening" for user input (as it showed on the console of R. When anyone who opens the dashboard selects a specific filter on the dashboard such as a year range or a specific technology type, then what happens is that the Shiny server instantly sends a new SQL query to the PostgreSQL backend part. The results then from the postgre backend are then processed and pushed back to the user interface, updating the visuals in milliseconds which is phenomenal once I see all of this working together! .

For hosting and deployment of the app, I utilized the shinyapps.io platform. By hosting the database and the dashboard in the cloud, it becomes possible for any researcher with a web link to access the tool without needing to install specialized software or manage their own database connections. This approach democratizes the data, allowing scholars from around the world to benefit from the structured Information Management principles applied in this project. Lastly, I used R studio, basically I connected my pgadmin to r an then once the tables were created in R i moved them to PG admin and then I connected my R to shiny which is connected to my github repository.

4. The User Interface (Shiny Dashboard)

While raw tables are important for verification, a dashboard is a UI/UX design tool which takes away cognitive load and makes it easy to reveal trends and patterns within the data and the tables of the database. Instead of scanning through thousands of rows of trade data from UN and WTO documents, the dashboard allows a user to immediately see global patterns, such as which regions are emerging as the primary importers of autonomous systems. This visual approach allows the researcher to grasp the "big picture" before diving into specific data points.

The dashboard features three core interactive components: SQL-backed filters, visualizations, and also a drill-down function. The filters for variables such as Year, Tech Type, and Region are directly tied to the database, which ensures that the interface is always a perfect reflection of the underlying data. Visualizations for these can be rendered using ggplot2 and plotly packages which are commonly used in R, turning dry numbers into interactive line graphs and heatmaps. These tools allow for an exploratory research style where the user can manipulate variables and see the results update in real-time.

The interactivity of the dashboard allows for the effective "drill-down" approach to data analysis. A user can start with a global view of all AI trades and then, through a series of clicks, filter down to a specific supplier and recipient to see the exact nature of their technology exchanges. This ability to be able to move seamlessly from a high-level summary to specific raw records is a direct result of the relational structure of the database, providing a level of transparency and detail that is essential for rigorous academic study.

5. Implementation Challenges & Lessons

One of the most significant challenges that I faced during implementation was managing referential integrity, it was particularly regarding the fact that the deletion and modification of parent records. I learned that in a relational system, you cannot simply delete a country or a manufacturer if they are tied to active trade records. This struggle for me was in implementing "On Delete Restrict" rules, which prevent the accidental deletion of data. This lesson underscored the importance of how once a relationship is established, the system has tot protect that link to ensure the history of the research remains intact.

Another key focus was designing for scalability. As I continue my PhD over the next year, my dataset will inevitably grow or i hope so. By building the database on relational principles from day one, I have been able to ensure that the as I keep working on this project it can handle thousands of additional records without needing a redesign. I can add new tables for more variables of "Conflict Events" or "Legislative Records" and simply link them to my existing master tables. This modularity ensures that the database will remain a functional asset throughout the entirety of my academic career and as I enter the job market.

Finally, the project provides a valuable lesson in balancing accessibility with security and the reliability of the data. While the goal is to make the data public, the database must remain protected from malicious and unreliable information or it may lead to accidental SQL injection. I address this by creating a read-only user account for the public Shiny dashboard. So on my dashboard one can only perform "Select" queries

7. Conclusion & Future Directions

In conclusion, the AI Tech Trade Database represents a significant step forward in the way we manage information within the field of International Relations specifically as it is a big part of my phd project. By applying the principles of relational design, normalization, and other database concepts that were taught, I have tried to transform a fragmented and unstructured problem of human rights research and data into a professional research tool. This database will serve as a beneficial side project for my PhD dissertation proposal, providing the high-quality, normalized data required to run advanced statistical models on conflict risk and civilian casualty thresholds.

Moving forward, I hope that I can transition this project from a classroom assignment into an actual resource for the broader Political Science community. Future iterations hopefully include the possibility of automated API feeds to keep the data current and an expanded schema to include domestic AI policy metrics. By providing a structured, relational view of the global tech trade, this project not only demonstrates a new understanding of Information Management concepts in the stream of international relations but also contributes a much-needed analytical framework to the study of technology, international law, and human rights and the interaction of these research interests.

AI DISCLOSURE

AI was used to sync the multiple applications as I was learning them and facing too many bugs. The AI tools that were taken in assistance were predominantly gemini.